IN A ONE-DIMENSIONAL NOETHERIAN LOCAL RING THE NUMBER OF MINIMAL PRIMES IS BOUNDED BY THE LENGTH OF THE QUOTIENT BY AN ELEMENT AVOIDING THEM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'as there is a surjection R/xR -> R'/xR" and concludes the length of the target is smaller; the conclusion as stated is the intended one and is reproduced unchanged, as is the phrase 'with minimal primes p_1, ..., p_t' following 'Assume R is reduced'.

Allowed standard prerequisites (editorial).

- Support of a finite module and finite length for a module supported at the closed point. used at proof lines 38048-38052. Shows Q has finite length and hence is annihilated by a power of x.
- The snake lemma applied to a square of multiplication maps. invoked at proof lines 38067-38070. Produces the exact sequence relating R/x^nR, M/x^nM and Q, which yields the length equality.
- The one-dimensional length lemma for the quotients R_i/xR_i. invoked at proof lines 38076-38077. Converts the length equality into the explicit alternating count from which the bound follows.

Source reference. Original lemma 00L6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 776–776. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07JW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9636–9636. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00L5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 775–775. Cross-chapter navigation pointer into the

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Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 37979: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37979–37979. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension 1. Let $x \in \mathfrak{m}$ be an element not contained in any minimal prime of R. Let t be the number of minimal prime ideals of R. Then $t \leq \text{length}_R(R/xR)$.*

Proof. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_t$ be the minimal prime ideals of R. Set $R' = R/\sqrt{0} = R/(\bigcap_{i=1}^t \mathfrak{p}_i)$. We claim it suffices to prove the lemma for R'. Namely, it is clear that R' has t minimal primes too and $\text{length}_{R'}(R'/xR') = \text{length}_R(R'/xR')$ is less than $\text{length}_R(R/xR)$ as there is a surjection $R/xR \to R'/xR'$. Thus we may assume R is reduced.

Assume R is reduced with minimal primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_t$. This means there is an exact sequence

$$0 \to R \to \prod_{i=1}^t R/\mathfrak{p}_i \to Q \to 0$$

Here Q is the cokernel of the first map. Write $M = \prod_{i=1}^t R/\mathfrak{p}_i$. Localizing at $\mathfrak{p}_j$ we see that

$$R_{\mathfrak{p}_j} \to M_{\mathfrak{p}_j} = \left(\prod_{i=1}^t R/\mathfrak{p}_i\right)_{\mathfrak{p}_j} = (R/\mathfrak{p}_j)_{\mathfrak{p}_j}$$

is surjective. Thus $Q_{\mathfrak{p}_j} = 0$ for all j. We conclude that $\text{Supp}(Q) = \{\mathfrak{m}\}$ as $\mathfrak{m}$ is the only prime of R different from the $\mathfrak{p}_i$. It follows that Q has finite length (Algebra, Lemma 00L5). Since $\text{Supp}(Q) = \{\mathfrak{m}\}$ we can pick an $n \gg 0$ such that x^n acts as 0 on Q (Algebra, Lemma 00L6). Now consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \longrightarrow & M & \longrightarrow & Q & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle x^n} & & \downarrow{\scriptstyle x^n} & & \downarrow{\scriptstyle x^n} & & \\ 0 & \longrightarrow & R & \longrightarrow & M & \longrightarrow & Q & \longrightarrow & 0 \end{array}$$

where the vertical maps are multiplication by x^n. This is injective on R and on M since x is not contained in any of the $\mathfrak{p}_i$. By the snake lemma (Algebra, Lemma 07JW), the following sequence is exact:

$$0 \to Q \to R/x^nR \to M/x^nM \to Q \to 0$$

Hence we find that $\text{length}_R(R/x^nR) = \text{length}_R(M/x^nM)$ for large enough n. Writing $R_i = R/\mathfrak{p}_i$ we see that $\text{length}(M/x^nM) = \sum_{i=1}^t \text{length}_R(R_i/x^nR_i)$. Applying Lemma 37979 and the fact that x is a nonzerodivisor on R and R_i, we conclude that

$$n\text{length}_R(R/xR) = \sum\nolimits_{i=1}^t n\text{length}_{R_i}(R_i/xR_i)$$

Since $\text{length}_{R_i}(R_i/xR_i) \geq 1$ the lemma is proved. □